

TANVIR HOSSAIN

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SUMMARY

Offensive hardware security researcher (Ph.D. expected 2026) with 4+ years of hands-on experience attacking silicon-level defenses: power/EM side-channel analysis, fault injection, hardware Trojan design, and gate-level netlist reverse engineering. Track record of published attacks and security tooling at IACR TCHES, ACM TODAES, and GLSVLSI; Champion of the IEEE HOST 2023 Microelectronics Security Challenge (IP security). Combines RTL/ASIC design-flow fluency (Verilog, Cadence and Xilinx flows) with low-level C, LLVM-based instrumentation, and machine-learning-driven attack automation. Currently building ML-guided fault-injection workflows at Keysight Technologies.

TECHNICAL SKILLS

Offensive Hardware Security:	Power/EM side-channel analysis (SCA) of cryptographic implementations, fault injection, hardware Trojan design and insertion, attacks on PUFs and obfuscated scan chains, gate-level netlist reverse engineering, threat modeling of design-for-security primitives.
ML for Security:	ML-assisted side-channel analysis, golden-chip-free hardware Trojan detection and localization, model-guided fault-injection parameter optimization.
Languages:	C (low-level/embedded), Python, Verilog, VHDL, Assembly (MIPS), Tcl, Bash, MATLAB; LLVM compiler-pass development.
SCA & Lab Equipment:	ChipWhisperer Husky; Keysight/Riscure Spyder, Inspector, and Pinata; high-precision EM probes and probe station; PicoScope; Tektronix DPO70404C oscilloscope; Analog Discovery 2; embedded-target bring-up and debugging.
Digital Design & EDA:	RTL design and verification; Cadence Genus, Encounter, Virtuoso; Xilinx Vivado, ISE; Intel Quartus; ModelSim; HSPICE, PSpice; gem5, Intel VTune, McPAT.
Hardware Platforms:	Xilinx Artix-7 (Basys 3, Nexys A7-100T), Intel MAX10 (HaHa Board), Altera DE2-115 FPGAs; STM32, ESP32, Arduino, Raspberry Pi.

EXPERIENCE

Keysight Technologies

May 2026 – Aug 2026

Machine Learning Security Intern — Applied ML & Fault Injection

Santa Rosa, CA

- Developing model-guided optimization and adaptive-learning workflows that automatically tune fault-injection parameters, targeting higher vulnerability-detection coverage than baseline sweep-based flows.
- Designing experiments and telemetry instrumentation; building scalable hyperparameter search and reliability metrics; delivering a prototype demonstration on production device-security test tooling.

The University of Kansas

Jan 2022 – Present

Graduate Research Assistant, Hardware Security (Advisor: Dr. Tamzidul Hoque)

Lawrence, KS

- Designed Trojan-assisted meta-attacks that systematically locate and neutralize design-for-security primitives in microelectronic designs; demonstrated authentication bypass of ring-oscillator PUFs and unlocking of dynamically obfuscated scan chains (ACM TODAES 2026).
- Led an in-depth security evaluation of lightweight homomorphic obfuscation schemes that protect software secrets against hardware Trojans (IACR TCHES 2026).
- Built SPHINX, an automated framework for identifying and extracting security primitives from gate-level netlists — reverse-engineering tooling that maps a design’s defensive surface for attack planning (GLSVLSI 2026).
- Developed golden-chip-free, ML-assisted power side-channel techniques for detecting and localizing hardware Trojans on FPGAs (Journal of Hardware and Systems Security 2026; IEEE NAECON 2023).
- Built HOACS, a software-only mitigation that keeps secrets encoded during execution on untrusted COTS processors via residue number coding, implemented as C-library and LLVM-based program transformations for cryptographic and arithmetic workloads.
- Proposed a gray-box, pre-deployment security-evaluation methodology for opaque COTS hardware, driven by power and EM side-channel measurements under zero-trust supply-chain assumptions.
- Demonstrated power side-channel and fault attacks on the ASCON lightweight-cryptography standard and evaluated counter-measures; delivered hands-on power-SCA tutorials and workshops.
- Mentored four NSF REU undergraduate researchers across two universities; co-developed and co-taught a hands-on computer hardware course using FPGAs and microcontrollers (NSF-funded).

EDUCATION

Ph.D. in Electrical Engineering (Hardware Security) , The University of Kansas	Expected 2026
Dissertation research: offensive hardware security — Trojan-assisted attacks on design-for-security primitives, side-channel analysis, and ML-assisted vulnerability detection.	
M.S. in Electrical Engineering , The University of Kansas	May 2025
B.Sc. in Electrical and Electronic Engineering , Ahsanullah University of Science and Technology	2016 – 2021

SELECTED PUBLICATIONS

17 peer-reviewed publications (7 journal articles, 10 conference papers); full list on [Google Scholar](#).

1. **Tanvir Hossain**, Matthew Showers, Mahmudul Hasan, and Tamzidul Hoque, “On the Security of Lightweight Homomorphic Obfuscation for Protecting Against Hardware Trojans,” *IACR Transactions on Cryptographic Hardware and Embedded Systems (TCHES)*, vol. 2026, no. 2, pp. 736–763, 2026. DOI: [10.46586/tches.v2026.i2.736-763](#)
2. **Tanvir Hossain**, S. M. Mojahidul Ahsan, Prabuddha Chakraborty, Swarup Bhunia, and Tamzidul Hoque, “Attacking the Protections of Hardware using Trojan-Assisted Meta Attacks,” *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 2026. DOI: [10.1145/3804452](#)
3. **Tanvir Hossain**, S. M. Mojahidul Ahsan, and Tamzidul Hoque, “SPHINX: A Framework for Security Primitive Hardware Identification and Extraction,” *Great Lakes Symposium on VLSI (GLSVLSI)*, 2026. DOI: [10.1145/3787109.3815307](#)
4. **Tanvir Hossain**, Ashutosh Ghimire (joint first author), Fathi Amsaad, Tamzidul Hoque, and Ahmed Sherif, “AI-Assisted Side-Channel Analysis for Golden-Chip Free Detection and Localization of Hardware Trojans,” *Journal of Hardware and Systems Security*, 2026. DOI: [10.1007/s41635-026-00182-4](#)
5. Ashutosh Ghimire, Fathi Amsaad, **Tanvir Hossain**, Tamzidul Hoque, and Ahmed Sherif, “FPGA Hardware Trojan Detection: Golden-Free Machine Learning Approach,” *IEEE National Aerospace and Electronics Conference (NAECON)*, pp. 181–186, 2023. DOI: [10.1109/NAECON58068.2023.10365812](#)
6. **Tanvir Hossain**, S. M. Mojahidul Ahsan, and Tamzidul Hoque, “Potential and Pitfalls of Multi-valued Logic Circuits for Hardware Security,” *IEEE Dallas Circuits and Systems Conference (DCAS)*, pp. 1–6, 2023. DOI: [10.1109/DCAS57389.2023.10130261](#)
7. S. M. Mojahidul Ahsan, Muhammad Sakib Shahriar, Mrityika Chowdhury, **Tanvir Hossain**, Md Sakib Hasan, and Tamzidul Hoque, “Accurate, Yet Scalable: A SPICE-based Design and Optimization Framework for eNVM based Analog In-memory Computing,” *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, pp. 1–9, 2024. DOI: [10.1145/3676536.3676827](#)

SELECTED HONORS & AWARDS

MAY 2023	Champion, Microelectronics Security Challenge (IP Security) — IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2023): first place in the IP-security track of HOST’s hardware security competition.
2022, ’23, ’26	NSF Travel Grants — IEEE HOST Symposium (awarded three times).
JANUARY 2024	First Place, Poster Presentation — I2S Student Research Symposium, University of Kansas.
APRIL 2023	David D. and Mildred H. Robb Award — EECS Department, University of Kansas.
MARCH 2021	Best Paper Award — 23rd International Conference on Computer and Information Technology (IC-CIT 2020).